

A discrete Plateau problem: soap films between two rings

by gradient descent on triangulated surfaces

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Abstract

We solve the classical Plateau problem for two coaxial rings numerically, by gradient descent on the area of a triangle mesh with fixed boundary. The key structural fact is that the vertex-wise gradient of discrete area is the integrated mean-curvature vector, so the descent is a discrete mean curvature flow: the mesh evolves the way a soap film drains area. With ~ 1300 vertices and pure `numpy`, the relaxed film reproduces the stable catenoid $r = c \cosh(z/c)$ to 0.09% in both area and neck radius, the sweep of final neck radii traces the stable branch $c(h)$ of the boundary-value problem to $\leq 0.6\%$, and past the critical separation $h^* \approx 1.3255 R$ every run pinches off toward the Goldschmidt configuration — so the numerics recovers, from local area minimization alone, the saddle-node bifurcation at which the catenoid ceases to exist.

1 Introduction

Plateau’s problem — find a surface of least area spanning a given boundary curve — sits at the historical center of the calculus of variations and geometric analysis, from Lagrange’s derivation of the minimal surface equation through Plateau’s soap-film experiments [3] to the existence theory of Douglas and Radó [5, 4]. The two-ring configuration is its oldest and most instructive instance: the minimal surfaces of revolution are catenoids, known since Euler [1], and yet the boundary-value problem already exhibits non-existence, non-uniqueness, and a genuine bifurcation. When the rings are pulled apart, the spanning film does not deform gracefully into something thinner — at a critical separation it ceases to exist at all, and a laboratory film snaps to the disconnected *Goldschmidt solution*, two flat disks [2].

This report documents a small, fully local numerical experiment: discretize the film as a triangle mesh with Dirichlet (wire) boundary, flow it by the exact gradient of discrete area, and confront the results with the closed-form catenoid theory. Everything runs in a few minutes on one CPU core with `numpy/scipy/matplotlib` only; the entire pipeline (solver, validation, figures, animation) is a single script, `plateau_solver.py`. The experiment is deliberately in the spirit of Brakke’s Surface Evolver [6]: geometry first, no black boxes, every number checkable against a theorem.

2 The catenoid and its fold

2.1 Minimal surfaces of revolution

Fix rings of radius R in the planes $z = \pm h/2$. For a surface of revolution with profile $r(z) > 0$ the area is $A[r] = 2\pi \int_{-h/2}^{h/2} r \sqrt{1 + r'^2} dz$. The integrand does not depend on z explicitly, so the

Euler–Lagrange equation has the Beltrami first integral $r/\sqrt{1+r'^2} = c$, giving the catenary profile

$$r(z) = c \cosh \frac{z - z_0}{c}, \quad \text{and by symmetry } z_0 = 0. \quad (1)$$

The boundary condition $r(\pm h/2) = R$ becomes a scalar equation for the neck parameter c ,

$$f(c) := c \cosh \frac{h}{2c} = R. \quad (2)$$

Since $f(c) \rightarrow \infty$ as $c \rightarrow 0^+$ and $f(R) > R$, with a single interior minimum, (2) has two roots $c_- < c_+$ for small h , one at tangency, and none beyond. Along the upper branch c_+ the catenoid is a local minimizer of area; the lower branch c_- is a saddle of the area functional with one unstable direction [9]. The area of the catenoid is obtained from $r\sqrt{1+r'^2} = c \cosh^2(z/c)$:

$$A_{\text{cat}}(c, h) = \pi c \left(h + c \sinh \frac{h}{c} \right). \quad (3)$$

2.2 The critical separation and the Goldschmidt solution

The two branches merge where (2) is tangent, i.e. where $f'(c) = \cosh \mu - \mu \sinh \mu = 0$ with $\mu := h/2c$:

$$\mu \tanh \mu = 1, \quad \mu \approx 1.199679, \quad \frac{h^*}{R} = \frac{2\mu}{\cosh \mu} \approx 1.325487, \quad \frac{c^*}{R} = \frac{1}{\cosh \mu} \approx 0.552434. \quad (4)$$

This is a saddle-node (fold) bifurcation: at $h = h^*$ the stable catenoid and the saddle collide and annihilate, and for $h > h^*$ *no* connected minimal surface spans the rings. The neck radius does not shrink to zero as the fold is approached — it stops at $c^* \approx 0.55 R$ and then the film must jump. What it jumps to is the Goldschmidt solution [2]: the degenerate competitor consisting of the two flat disks, of total area $2\pi R^2$, which exists for every h . Comparing (3) with $2\pi R^2$ shows the catenoid is the *global* minimizer only for $h \lesssim 1.0554 R$; on $1.0554 \lesssim h/R < 1.3255$ it is merely metastable — exactly like a laboratory soap film, which persists on the catenoid branch until the branch ceases to exist.

2.3 Area descent is mean curvature flow

For a smooth surface Σ with fixed boundary, the first variation of area along a normal field X is $\delta A(X) = -\int_{\Sigma} \mathbf{H} \cdot X$, where $\mathbf{H}$ is the mean-curvature vector. The L^2 -gradient flow of area is therefore mean curvature flow (MCF), $\partial_t x = \mathbf{H}$, the standard model for the over-damped relaxation of an interface driven by surface tension. Stationary points with the given wire boundary are exactly the minimal surfaces we seek, and the flow can only ever decrease area. The discrete experiment below is the literal transcription of this picture to a triangle mesh.

3 Discretization and algorithm

Mesh. The initial surface is the straight cylinder $r = R$, triangulated on an $n_z \times n_\theta$ vertex grid, periodic in θ , with the two extreme rows clamped to the rings ($n_\theta = 48$; n_z odd and scaled with h so the axial spacing is $\approx 0.04 R$). At $h = R$ this gives 1296 vertices and 2496 triangles. All experiments set $R = 1$.

The exact area gradient. Discrete area is $A(V) = \sum_T \text{area}(T)$ over triangles $T = (a, b, c) \subset \mathbb{R}^3$.

Lemma 1. *Let $n = (b - a) \times (c - a)$, $\hat{n} = n/|n|$. Then $\nabla_a \text{area}(T) = \frac{1}{2} \hat{n} \times (c - b)$, and cyclically for b, c .*

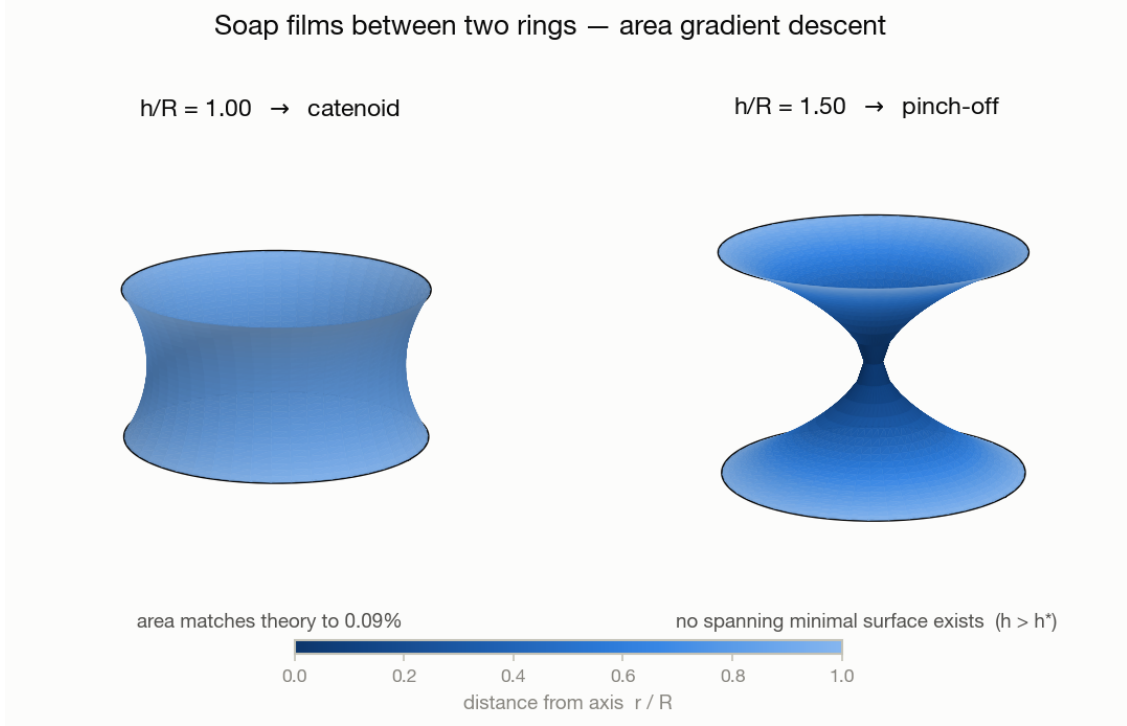


Figure 1: Final states of the flow, faces coloured by distance from the axis. Left: at $h/R = 1$ the mesh has converged to the catenoid; its area matches (3) to 0.09%. Right: at $h/R = 1.5 > h^*/R$ no spanning minimal surface exists and the film pinches off toward the Goldschmidt configuration.

Proof. $\text{area} = \frac{1}{2}|n|$, so $d \text{ area} = \frac{1}{2}\hat{n} \cdot dn$. Varying a only, $dn = (-da) \times (c - a) + (b - a) \times (-da) = da \times (b - c)$, hence $d \text{ area} = \frac{1}{2}\hat{n} \cdot (da \times (b - c)) = \frac{1}{2}((b - c) \times \hat{n}) \cdot da$. $\square$

The gradient at a vertex is thus half the opposite edge, rotated by 90° in the triangle plane — and summing it over the triangles incident to a vertex yields precisely the cotangent-formula discretization of the integrated mean-curvature vector [7, 8]. Dividing by the lumped (barycentric) vertex mass $m_v = \frac{1}{3} \sum_{T \ni v} \text{area}(T)$ turns the area gradient into a pointwise mean-curvature velocity, and explicit Euler,

$$V \leftarrow V - \Delta t \frac{\nabla A(V)}{m}, \quad \Delta t = 0.2 \ell_{\min}^2, \quad (5)$$

is a discrete MCF. The heat-equation step size (5) ($\ell_{\min}$ = shortest edge, $\Delta t \approx 3 \times 10^{-4}$ at $h = R$) is refreshed every 25 iterations as the mesh deforms, and a per-step displacement cap of $\frac{1}{2}\ell_{\min}$ keeps the pinch-off regime tame. Each iteration is $O(\#\text{triangles})$, fully vectorized (the scatter-sums are `np.bincount` calls).

Termination. The flow has three exits: *converged* (relative area decrease below 10^{-11} for 30 consecutive steps), *collapsed* (neck radius $\min_i \sqrt{x_i^2 + y_i^2} < 0.06 R$, only possible past the fold, where the film is pinching), or an iteration cap. A capped run whose neck is still moving — the signature of critical slowing-down near h^* — is continued from its current state with a larger budget rather than misclassified. The full experiment (two instrumented runs, a 12-value sweep across the fold, all figures and the animation) takes 2–3 minutes on one CPU core, deterministically.

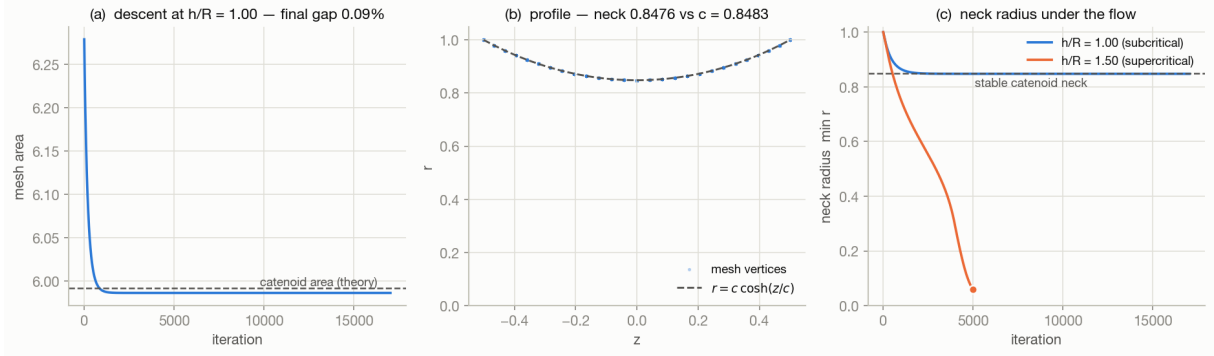


Figure 2: Validation at $h/R = 1$. (a) Area along the descent against the exact catenoid area (3). (b) All mesh vertices (z_i, r_i) of the relaxed film against the catenary profile. (c) Neck radius along the flow: subcritical (blue) settles at the stable c ; supercritical (orange, $h/R = 1.5$) never stabilizes and crashes to the pinch-off detection threshold.

4 Results

4.1 Subcritical separation: convergence to the catenoid

At $h = R$ the stable root of (2) is $c = 0.848338$, with $A_{\text{cat}} = 5.991797$. Starting from the straight cylinder (area $2\pi Rh = 6.283$), the flow converges after 17 038 iterations (flow time ≈ 5 in units of R^2) to a mesh of area 5.986563 and neck radius 0.847551: both observables agree with the exact catenoid to 0.09%, which is the expected $O(\ell^2)$ discretization accuracy of the mesh, not a limitation of the flow. Figure 1 (left) shows the converged film; Figure 2 shows the monotone area descent onto the theoretical value (a), the relaxed vertex cloud landing on the catenary $r = c \cosh(z/c)$ with no visible scatter (b), and the neck radius settling at c (c, blue).

4.2 Supercritical separation: pinch-off

At $h = 1.5 R > h^*$ the same flow has no stationary state to find. The neck shrinks monotonically, passes the fold value c^* without pausing, and accelerates into a pinch; the run terminates at the collapse threshold after 4 974 iterations (Figure 1, right; orange curve in Figure 2c). Since the mesh has fixed cylindrical connectivity, the flow cannot perform the topological transition to the two disks — it terminates at the incipient singularity instead, which is exactly the Goldschmidt limit seen from within the smooth category. The animation `film_evolution.gif` shows the two runs side by side.

4.3 Sweep across the fold: the bifurcation diagram

Repeating the experiment for twelve separations gives Table 1 and the bifurcation diagram of Figure 3. Every subcritical run lands on the stable branch $c(h)$, with deviation growing from 0.01% far from the fold to 0.55% at $h/R = 1.30$ — the energy landscape flattens as the stable catenoid and the saddle approach each other, so both the discrete equilibrium and the flow's convergence degrade together. Every supercritical run, including the delicate $h/R = 1.35$ just past the fold (which required the continuation described in §3), pinches off. The flow also tracks the catenoid branch far beyond the global-optimality crossover $h \approx 1.0554 R$: local gradient descent knows nothing about the lower-area Goldschmidt competitor, reproducing the metastability (hysteresis) of physical films.

Table 1: Final neck radius of the flow against the stable catenoid parameter $c(h)$ solving (2), for $R = 1$. Above the fold no catenoid exists and the flow pinches off.

h/R	neck (flow)	$c(h)$	deviation	outcome
0.50	0.9674	0.9675	0.01%	converged
0.70	0.9333	0.9336	0.03%	converged
0.90	0.8822	0.8828	0.07%	converged
1.00	0.8476	0.8483	0.09%	converged
1.10	0.8035	0.8046	0.14%	converged
1.20	0.7434	0.7451	0.23%	converged
1.25	0.7014	0.7037	0.33%	converged
1.30	0.6381	0.6416	0.55%	converged
1.35	—	—	—	pinch-off
1.40	—	—	—	pinch-off
1.50	—	—	—	pinch-off
1.60	—	—	—	pinch-off

5 Discussion

Three features of the experiment deserve emphasis.

The right gradient makes the physics automatic. No curvature was ever computed: the scheme only differentiates the elementary formula $\text{area} = \frac{1}{2}|(b-a) \times (c-a)|$. That this descent is mean curvature flow — and therefore inherits its qualitative theory: monotone area, convergence to stable minimal surfaces, finite-time neck singularities — is the discrete shadow of the first-variation formula, via the identity between the summed area gradient and the cotangent Laplacian applied to the coordinates [7, 8].

The bifurcation is visible in the dynamics, not just the statics. Near h^* the spectral gap of the stability operator closes, and the flow exhibits textbook critical slowing-down: at $h/R = 1.35$ the neck drifts slowly for thousands of iterations before the pinch accelerates, while at $h/R = 1.5$ collapse is fast. Conversely the growth of the deviation column in Table 1 toward the fold is the static face of the same flattening landscape.

What the discrete flow cannot do is also instructive. With fixed connectivity the flow stops at the incipient pinch instead of continuing past the singularity to the two disks; continuing through topology change is precisely what measure-theoretic weak formulations (Brakke flow) and remeshing-based tools such as the Surface Evolver [6] are built for. Our collapse detector is thus an honest declaration of the model’s boundary, not a numerical failure — and its firing point coincides with the theoretical non-existence region to within one grid point of the sweep.

6 Outlook

The same forty lines of core solver generalize in directions that map onto the later units of this course: replacing the ring boundary by a disk-topology mesh over an Enneper-type wire frame probes non-uniqueness for a genuinely three-dimensional boundary; swapping explicit descent for the Pinkall–Polthier iteration [7] (solve $\Delta_{\Sigma}x = 0$ with cotangent weights per step) trades many cheap steps for few linear solves; and replacing area by the Willmore energy $\int H^2$ changes the flow’s character entirely. Finally, the converged meshes and the closed-form branch $c(h)$ constitute a compact, exactly-labelled dataset against which learned solvers — e.g. physics-informed networks for the minimal surface equation — can be benchmarked in later projects.

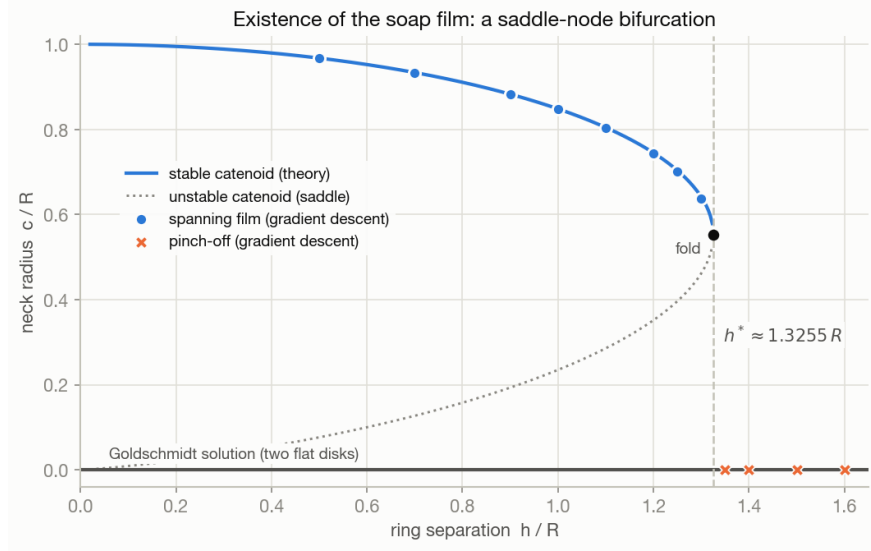


Figure 3: The saddle-node bifurcation of the two-ring soap film. Solid: stable catenoid branch; dotted: unstable saddle branch; the branches merge at the fold (4) and annihilate. Markers: final neck radius of each gradient descent run — on the stable branch below h^* , at zero (pinch-off, Goldschmidt) above it.

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